Lesson 16: Normed Spaces and Banach Spaces

Curriculum: A Rigorous Learning Path to Multi-Agent Deep RL

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3 Normed Spaces

3.1 Definition and Basic Properties

Definition 3.1 (Normed Space)**.** A *normed space* (or *normed vector space*) is a pair $(X, \|\cdot\|)$ where X is a vector space over $\mathbb{R}$ (or $\mathbb{C}$) and $\|\cdot\|\colon X \to [0,\infty)$ is a *norm*, i.e., for all $x, y \in X$ and α scalar:

(N1) $\|x\| = 0$ if and only if $x = 0$. (positive definiteness)
(N2) $\|\alpha x\| = |\alpha|\,\|x\|$. (absolute homogeneity)
(N3) $\|x + y\| \le \|x\| + \|y\|$. (triangle inequality)

Proposition 3.1 (Every Norm Induces a Metric)**.** *If $(X, \|\cdot\|)$ is a normed space, then $d(x, y) = \|x - y\|$ defines a metric on X. This metric is* translation-invariant *($d(x + z, y + z) = d(x, y)$) and* homogeneous *($d(\alpha x, \alpha y) = |\alpha| d(x, y)$).*

Proof. **??**: $d(x, y) = \|x - y\| = 0 \iff x - y = 0 \iff x = y$. **??**: $d(x, y) = \|x - y\| = \|-(y - x)\| = |-1| \cdot \|y - x\| = d(y, x)$. **??**: $d(x, z) = \|x - z\| = \|(x - y) + (y - z)\| \le \|x-y\|+\|y-z\| = d(x, y)+d(y, z)$. Translation invariance: $d(x+z, y+z) = \|(x+z)-(y+z)\| = \|x - y\| = d(x, y)$. Homogeneity: $d(\alpha x, \alpha y) = \|\alpha x - \alpha y\| = |\alpha|\|x - y\| = |\alpha| d(x, y)$. □

Remark 3.1. Not every metric comes from a norm. The discrete metric on $\mathbb{R}^n$ (with $n \ge 1$) is not translation-invariant (take $x = 0$, $y = z$ with $z \ne 0$: $d(x, y) = 1$ but $d(x+z, y+z) = d(z, 2z) = 1$, which happens to be invariant in this case, but it is not homogeneous: $d(2x, 2y) = 1 \ne 2 = 2d(x, y)$ when $x \ne y$). A metric arises from a norm if and only if it is translation-invariant and absolutely homogeneous.

3.2 Hölder's and Minkowski's Inequalities

To show that ℓ^p norms are indeed norms, we need the following fundamental inequalities.

Lemma 3.2 (Young's Inequality)**.** *For $a, b \ge 0$ and $p, q > 1$ with $\frac{1}{p} + \frac{1}{q} = 1$:*

$$ab \le \frac{a^p}{p} + \frac{b^q}{q}.$$

Proof. The function $\varphi(t) = e^t$ is convex. By the definition of convexity, $e^{\lambda s+(1-\lambda)t} \le \lambda e^s + (1 - \lambda)e^t$ for $\lambda \in [0, 1]$. Set $\lambda = 1/p$, $s = p \ln a$, $t = q \ln b$ (assuming $a, b > 0$):

$$ab = e^{\ln a+\ln b} = e^{(1/p)(p\ln a)+(1/q)(q\ln b)} \le \frac{1}{p}e^{p\ln a} + \frac{1}{q}e^{q\ln b} = \frac{a^p}{p} + \frac{b^q}{q}.$$

The case $a = 0$ or $b = 0$ is trivial. □

Proposition 3.3 (Hölder's Inequality)**.** *Let $1 < p < \infty$ and $q = p/(p - 1)$ (so $1/p + 1/q = 1$). For $x = (x_i), y = (y_i) \in \mathbb{R}^n$ (or sequences in ℓ^p, ℓ^q):*

$$\sum_i |x_i y_i| \le \|x\|_p \, \|y\|_q.$$

Proof. If $\|x\|_p = 0$ or $\|y\|_q = 0$, the result is trivial. Otherwise, set $a_i = |x_i|/\|x\|_p$ and $b_i = |y_i|/\|y\|_q$. By Young's inequality: $a_i b_i \le a_i^p/p + b_i^q/q$. Summing:

$$\sum_i a_i b_i \le \frac{1}{p}\sum_i \frac{|x_i|^p}{\|x\|_p^p} + \frac{1}{q}\sum_i \frac{|y_i|^q}{\|y\|_q^q} = \frac{1}{p} + \frac{1}{q} = 1.$$

Multiplying both sides by $\|x\|_p\|y\|_q$ gives the result. □

Proposition 3.4 (Minkowski's Inequality)**.** *For $1 \le p < \infty$ and $x, y \in \mathbb{R}^n$ (or ℓ^p): $\|x+y\|_p \le \|x\|_p + \|y\|_p$.*

Proof. The case $p = 1$ is the ordinary triangle inequality for absolute values. For $p > 1$, let $q = p/(p-1)$. We have:

$$\begin{aligned}\|x+y\|_p^p &= \sum_i |x_i + y_i|^p = \sum_i |x_i + y_i|^{p-1}|x_i + y_i| \\ &\le \sum_i |x_i + y_i|^{p-1}|x_i| + \sum_i |x_i + y_i|^{p-1}|y_i|.\end{aligned}$$

Apply Hölder's inequality to each sum with exponents q and p. Note that $(p-1)q = p$, so $\left(\sum_i |x_i + y_i|^{(p-1)q}\right)^{1/q} = \|x+y\|_p^{p/q}$. Therefore:

$$\|x+y\|_p^p \le \|x+y\|_p^{p/q}\big(\|x\|_p + \|y\|_p\big).$$

Dividing both sides by $\|x+y\|_p^{p/q}$ (if nonzero) and using $p - p/q = 1$ gives $\|x+y\|_p \le \|x\|_p + \|y\|_p$. □

3.3 Important Examples of Normed Spaces

Example 3.1 ($\mathbb{R}^n$ with p-norms)**.** For $1 \le p \le \infty$, $(\mathbb{R}^n, \|\cdot\|_p)$ is a normed space. Positive definiteness and homogeneity are immediate; the triangle inequality for $1 \le p < \infty$ is Minkowski's inequality. For $p = \infty$, $\|x+y\|_\infty = \max_i |x_i + y_i| \le \max_i(|x_i| + |y_i|) \le \max_i |x_i| + \max_i |y_i| = \|x\|_\infty + \|y\|_\infty$.

Example 3.2 ($C[a,b]$. $with Supremum Norm$]C[a,b]$with \|f\|_\infty = \max_{t\in[a,b]} |f(t)|$ is a normed space. The maximum exists because f is continuous on the compact set $[a,b]$.

Example 3.3 ($C[a,b]$. $with$L^1 Norm] $C[a,b]$ with $\|f\|_1 = \int_a^b |f(t)|\,dt$ is a normed space. This norm captures "average" rather than "worst-case" behavior.

Example 3.4 (Sequence Spaces ℓ^p)**.** For $1 \le p \le \infty$, $(\ell^p, \|\cdot\|_p)$ is a normed space by Minkowski's inequality (extended to infinite sums).

3.4 Equivalence of Norms in Finite Dimensions

Definition 3.2 (Equivalent Norms)**.** Two norms $\|\cdot\|_a$ and $\|\cdot\|_b$ on a vector space X are *equivalent* if there exist constants $c, C > 0$ such that

$$c\|x\|_a \le \|x\|_b \le C\|x\|_a \quad \text{for all } x \in X.$$

Remark 3.2. Equivalent norms define the same topology: they determine the same open sets, convergent sequences, Cauchy sequences, and continuous functions. In particular, a space is complete under one norm if and only if it is complete under any equivalent norm.

Theorem 3.5 (All Norms on $\mathbb{R}^n$ Are Equivalent)**.** *On $\mathbb{R}^n$, every two norms are equivalent.*

Proof. It suffices to show that any norm $\|\cdot\|$ on $\mathbb{R}^n$ is equivalent to $\|\cdot\|_\infty$, since equivalence is transitive.

Step 1: $\|\cdot\|$ is bounded above by a multiple of $\|\cdot\|_\infty$. Let $e_1, \ldots, e_n$ be the standard basis. For $x = \sum_{i=1}^n x_i e_i$:

$$\|x\| = \Big\|\sum_{i=1}^n x_i e_i\Big\| \le \sum_{i=1}^n |x_i|\,\|e_i\| \le \|x\|_\infty \sum_{i=1}^n \|e_i\| = C\|x\|_\infty,$$

where $C = \sum_{i=1}^{n} \|e_i\|$.

Step 2: $\|\cdot\|$ is continuous with respect to $\|\cdot\|_\infty$. By Step 1, $|\|x\| - \|y\|| \le \|x - y\| \le C\|x - y\|_\infty$, so $x \mapsto \|x\|$ is Lipschitz (hence continuous) on $(\mathbb{R}^n, \|\cdot\|_\infty)$.

Step 3: $\|\cdot\|$ is bounded below by a multiple of $\|\cdot\|_\infty$. The unit sphere $S = \{x \in \mathbb{R}^n : \|x\|_\infty = 1\}$ is closed and bounded in $(\mathbb{R}^n, \|\cdot\|_\infty)$, hence compact (by the Heine–Borel theorem). The continuous function $x \mapsto \|x\|$ attains its minimum on S: let $c = \min_{x \in S} \|x\|$. Since $0 \notin S$ (because $\|x\|_\infty = 1$ implies $x \neq 0$, so $\|x\| > 0$ by (N1)), we have $c > 0$. For any $x \neq 0$, $x/\|x\|_\infty \in S$, so $\|x/\|x\|_\infty\| \ge c$, giving $\|x\| \ge c\|x\|_\infty$.

Combining Steps 1 and 3: $c\|x\|_\infty \le \|x\| \le C\|x\|_\infty$. □

Finite-Dimensional Norm Equivalence

On a finite-dimensional vector space, all norms are equivalent. This means convergence, completeness, compactness, and continuity are norm-independent in $\mathbb{R}^n$. This fails dramatically in infinite dimensions (see Warning 3.4).

Corollary 3.6. *Every finite-dimensional normed space is complete (i.e., is a Banach space).*

Proof. Any norm on $\mathbb{R}^n$ is equivalent to $\|\cdot\|_\infty$, and $(\mathbb{R}^n, \|\cdot\|_\infty)$ is complete by Theorem **??**. Since equivalent norms have the same Cauchy sequences and convergent sequences, completeness is preserved. □

Infinite-Dimensional Spaces Are Different

On $C[0,1]$, the norms $\|f\|_\infty = \max |f|$ and $\|f\|_1 = \int_0^1 |f|$ are *not* equivalent. Consider $f_n(t) = t^n$: then $\|f_n\|_\infty = 1$ but $\|f_n\|_1 = 1/(n+1) \to 0$. If the norms were equivalent, $\|f_n\|_1 \to 0$ would imply $\|f_n\|_\infty \to 0$, a contradiction. This has profound consequences: $C[0,1]$ is complete under $\|\cdot\|_\infty$ but *not* under $\|\cdot\|_1$.

Norm Choice in RL

The choice of norm matters for the convergence properties of RL algorithms. Value iteration contracts in the $\|\cdot\|_\infty$ norm with factor γ, but the same operator need not contract in the $\|\cdot\|_2$ norm. For finite state spaces all norms are equivalent (Theorem 3.5), so convergence in one norm implies convergence in all. For infinite state spaces with function approximation, the norm matters critically, and non-equivalence of norms (Warning 3.4) is the root of many difficulties in approximate dynamic programming.

4 Banach Spaces

Definition 4.1 (Banach Space)**.** A *Banach space* is a complete normed space, i.e., a normed space $(X, \|\cdot\|)$ that is complete as a metric space under the induced metric $d(x, y) = \|x - y\|$.

4.1 Examples and Non-Examples

Theorem 4.1 (ℓ^p is a Banach Space)**.** *For $1 \le p \le \infty$, $(\ell^p, \|\cdot\|_p)$ is a Banach space.*

Proof. We prove the case $1 \le p < \infty$; the case $p = \infty$ is analogous. Let $(x^{(k)})_{k=1}^{\infty}$ be a Cauchy sequence in ℓ^p, where $x^{(k)} = (x_1^{(k)}, x_2^{(k)}, \ldots)$.

Step 1: Identify the candidate limit. For each fixed i, $|x_i^{(k)} - x_i^{(m)}| \le \|x^{(k)} - x^{(m)}\|_p \to 0$, so $(x_i^{(k)})_k$ is Cauchy in $\mathbb{R}$, hence convergent. Define $x_i = \lim_{k\to\infty} x_i^{(k)}$ and $x = (x_1, x_2, \ldots)$.

Step 2: Show $x \in \ell^p$. Fix $\varepsilon > 0$ and choose K such that $\|x^{(k)} - x^{(m)}\|_p < \varepsilon$ for $k, m \geq K$. For any finite N:

$$\Big(\sum_{i=1}^{N} |x_i^{(k)} - x_i^{(m)}|^p\Big)^{1/p} \leq \|x^{(k)} - x^{(m)}\|_p < \varepsilon.$$

Taking $m \to \infty$ (with N fixed): $\big(\sum_{i=1}^{N} |x_i^{(k)} - x_i|^p\big)^{1/p} \leq \varepsilon$. Since this holds for all N, $\|x^{(k)} - x\|_p \leq \varepsilon < \infty$. By the triangle inequality, $\|x\|_p \leq \|x - x^{(k)}\|_p + \|x^{(k)}\|_p < \infty$, so $x \in \ell^p$.

Step 3: Show $x^{(k)} \to x$ in ℓ^p. From Step 2, $\|x^{(k)} - x\|_p \leq \varepsilon$ for all $k \geq K$, so $x^{(k)} \to x$ in ℓ^p. □

Theorem 4.2 ($C[a,b]$. *isBanachUndertheSupNorm*]*(C[a,b],* $\|\cdot\|_\infty$) *is a Banach space.*

Proof. Let (f_n) be Cauchy in $(C[a,b], \|\cdot\|_\infty)$. For each fixed $t \in [a,b]$, $|f_n(t) - f_m(t)| \leq \|f_n - f_m\|_\infty \to 0$, so $(f_n(t))$ is Cauchy in $\mathbb{R}$, hence convergent. Define $f(t) = \lim_{n\to\infty} f_n(t)$.

Uniform convergence. Given $\varepsilon > 0$, choose N such that $\|f_n - f_m\|_\infty < \varepsilon$ for $n, m \geq N$. For each t, $|f_n(t) - f_m(t)| < \varepsilon$. Letting $m \to \infty$: $|f_n(t) - f(t)| \leq \varepsilon$ for all t and $n \geq N$. Hence $\|f_n - f\|_\infty \leq \varepsilon$, so $f_n \to f$ uniformly.

Continuity of f. The uniform limit of continuous functions is continuous: for any $t_0 \in [a,b]$ and $\varepsilon > 0$, choose n with $\|f_n - f\|_\infty < \varepsilon/3$, then $\delta > 0$ with $|f_n(t) - f_n(t_0)| < \varepsilon/3$ for $|t - t_0| < \delta$. Then $|f(t) - f(t_0)| \leq |f(t) - f_n(t)| + |f_n(t) - f_n(t_0)| + |f_n(t_0) - f(t_0)| < \varepsilon$.

So $f \in C[a,b]$ and $f_n \to f$ in $\|\cdot\|_\infty$. □

Example 4.1 ($C[a,b]$. *IsNotBanachUnder*$\|\cdot\|_1$] The space $(C[0,1], \|\cdot\|_1)$ is *not* complete. Define $f_n\colon [0,1] \to \mathbb{R}$ by

$$f_n(t) = \begin{cases} 0 & \text{if } 0 \leq t \leq 1/2, \\ n(t - 1/2) & \text{if } 1/2 < t \leq 1/2 + 1/n, \\ 1 & \text{if } 1/2 + 1/n < t \leq 1. \end{cases}$$

Each f_n is continuous, and (f_n) is Cauchy under $\|\cdot\|_1$ (since $\|f_n - f_m\|_1 \leq 1/\min(m,n) \to 0$). But the pointwise limit is the discontinuous step function $\mathbf{1}_{(1/2,1]}$, which is not in $C[0,1]$. No continuous function can be the $\|\cdot\|_1$-limit either, since $\|f_n - g\|_1 \to 0$ and $\|f_n - \mathbf{1}_{(1/2,1]}\|_1 \to 0$ would give $g = \mathbf{1}_{(1/2,1]}$ a.e., contradicting continuity of g.

4.2 Series in Banach Spaces

Definition 4.2 (Absolute Convergence)**.** A series $\sum_{n=1}^{\infty} x_n$ in a normed space is *absolutely convergent* if $\sum_{n=1}^{\infty} \|x_n\| < \infty$. It is *(unconditionally) convergent* if the partial sums $S_N = \sum_{n=1}^{N} x_n$ converge in norm.

Theorem 4.3 (Characterization of Banach Spaces via Series)**.** *A normed space X is a Banach space if and only if every absolutely convergent series in X converges.*

Proof. ($\Rightarrow$) Suppose X is Banach and $\sum \|x_n\| < \infty$. The partial sums $S_N = \sum_{n=1}^{N} x_n$ form a Cauchy sequence: for $M > N$,

$$\|S_M - S_N\| = \Big\| \sum_{n=N+1}^{M} x_n \Big\| \leq \sum_{n=N+1}^{M} \|x_n\| \to 0$$

as $N, M \to \infty$ (tail of a convergent series). Since X is complete, (S_N) converges.

($\Leftarrow$) Suppose every absolutely convergent series converges. Let (y_n) be a Cauchy sequence. Choose a subsequence (y_{n_k}) with $\|y_{n_{k+1}} - y_{n_k}\| < 2^{-k}$ (possible since (y_n) is Cauchy). Define $x_1 = y_{n_1}$ and $x_k = y_{n_k} - y_{n_{k-1}}$ for $k \geq 2$. Then the partial sums satisfy $\sum_{k=1}^{K} x_k = y_{n_K}$,

and $\sum_{k=2}^{\infty} \|x_k\| \leq \sum_{k=1}^{\infty} 2^{-k} = 1 < \infty$. So $\sum x_k$ converges absolutely, hence converges by hypothesis: $y_{n_K} \to y$ for some $y \in X$. Since (y_n) is Cauchy and has a convergent subsequence, the full sequence converges to y (by the argument: $\|y_n - y\| \leq \|y_n - y_{n_K}\| + \|y_{n_K} - y\| \to 0$ for suitable K). □

Banach Space Characterization

A normed space is Banach if and only if every absolutely convergent series converges. This characterization is often the most convenient way to verify completeness: one constructs a convergent series rather than chasing arbitrary Cauchy sequences.

Why Completeness Matters for RL

The Banach fixed-point theorem (proved in the next lesson) states: if T is a contraction on a *complete* metric space, then T has a unique fixed point and $T^n x \to x^*$ for any starting point x. The Bellman operator is a γ-contraction on the Banach space of bounded functions with the sup norm. Without completeness, we cannot guarantee that the sequence of value iterates converges to anything in our space. This is the mathematical reason we work in Banach spaces, not just normed spaces.

4.3 Schauder Basis

Definition 4.3 (Schauder Basis)**.** A sequence $(e_n)_{n=1}^{\infty}$ in a Banach space X is a *Schauder basis* if every $x \in X$ has a unique representation $x = \sum_{n=1}^{\infty} \alpha_n e_n$ (convergent in norm).

Example 4.2. The standard unit vectors (e_n) form a Schauder basis for ℓ^p $(1 \leq p < \infty)$: every $x = (x_1, x_2, \ldots) \in \ell^p$ satisfies $x = \sum_{n=1}^{\infty} x_n e_n$ in $\|\cdot\|_p$. Note that (e_n) is *not* a Schauder basis for ℓ^∞, because the partial sums $\sum_{n=1}^{N} x_n e_n$ need not converge to x in $\|\cdot\|_\infty$.